

ESPRIT · 2026

FINAL YEAR PROJECT

Wayloom.AI

An AI-Powered Cognitive Testing Platform

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1 INTRODUCTION & CONTEXT

Presentation Overview

01 Context & Problem

02 Solution & Demo

03 Architecture

04 AI Engineering

05 Verification & Roadmap

06 Conclusion

2 COMPANY PRESENTATION

Prospecter — Internship Host Company

AI-powered B2B platform for outbound prospecting.

MISSION

Autonomous AI SDRs help B2B teams scale outbound prospecting.

AI / LLM

B2B SAAS

SALES AUTOMATION

OUTBOUND GTM



AI SDR Engine



Lead Intelligence



Outbound Automation



Pipeline Analytics



Strategic context — LLM workflows · Multi-tenancy · Background workers

2 INTERNSHIP CONTEXT

Internship Context

Software Engineering internship at **Prospecter**, an AI-powered B2B platform.



Full-Stack Engineering



AI Integration



Platform Engineering

Scope: Independent product · Production-inspired engineering

SOFTWARE ENGINEER

AI INTEGRATION

SOLO CONTRIBUTOR

3 CLINICAL CONTEXT

Cognitive Health in the Digital Era



Aging population



Clinical overload



Caregiver expectations



AI adoption in healthcare

3 CLINICAL CONTEXT

Dementia is increasing, testing isn't

+6.9M

Americans 65+ living with dementia
2024, projected to reach 12M by 2040

50%

Never receive a clinical diagnosis
of those affected

2x

Rates double every 5 years after 65
WHO public-health priority



Paper testing since the 1980s



Long clinic visits & wait times



Months-long waitlists



Clinical overload

Cost of delay: half of those affected **never receive a clinical diagnosis** before it's too late to act.

Engineering response: digitize testing, keep it accessible, and get patients seen faster.

4 EXISTING SOLUTIONS & GAP

Competitive Landscape

How existing cognitive-testing tools cover adaptation, delivery, and caregiver access.

FEATURE	BRAINCHECK	LINUS HEALTH	ALTOIDA	WAYLOOM
Cultural Adaptation	✗	✗	✗	✓
Adaptive Testing	✗	✗	✗	✓
Conversational Delivery	✗	✗	✗	✓
Longitudinal Monitoring	✓	✓	✓	✓
Caregiver Integration	✗	✗	✗	✓
At-Home Usability	✗	✗	✗	✓

Observation: existing tools monitor cognitive decline longitudinally, but none combine cultural adaptation, adaptive conversational testing, and caregiver integration in one at-home-usable product.

4 EXISTING SOLUTIONS & GAP

An exploding market for cognitive health



Rapid growth: >25% CAGR over 6 years



Large TAM: \$31M global market by 2030



Underserved US: \$2B market, few accessible tools

Accelerating dementia

Clinical overload

Policy & tech momentum



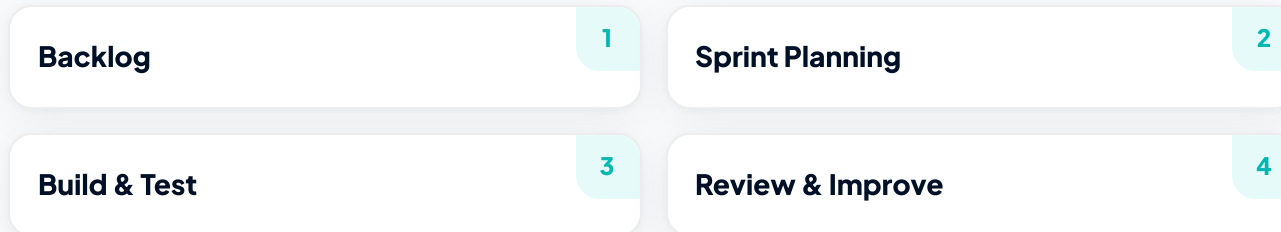
This gap defines the project's objective

One accessible, culturally-adaptive cognitive-testing platform.

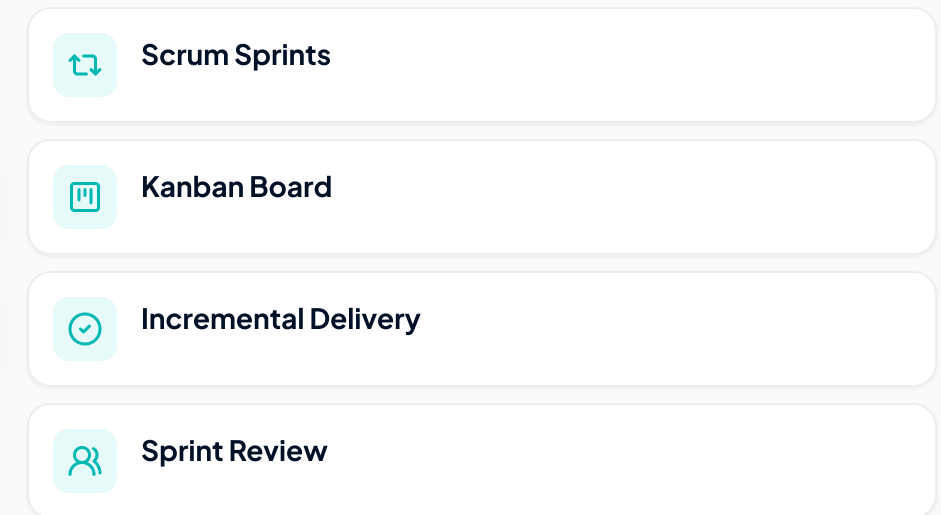
5 METHODOLOGY & REQUIREMENTS

Engineering Methodology

SPRINT CYCLE



HOW WORK WAS MANAGED



5 FUNCTIONAL REQUIREMENTS

Core platform requirements

Core platform capabilities for institutions, clinicians, and patients.



FR-01

Authentication & Security

- Argon2 password protection
- JWT + OAuth sign-in



FR-02

Multi-Tenant Architecture

- Business-scoped records
- Tenant context enforced



FR-03

Roles & Permissions

- Role and plan guards
- Feature access by subscription



FR-04

Test Digitization & Management

- Upload or select validated cognitive tests
- Define scoring rules & normative data



FR-05

Patient Session Pipeline

- Consent, intake & adaptive delivery
- Real-time AI scoring & clinician report

5 FUNCTIONAL REQUIREMENTS

AI requirements

Implemented intelligence capabilities in the AI Brain pipeline.



FR-06

Document Digitization

- Vision + OCR page understanding
- Structured items from paper tests



FR-07

AI-Assisted Scoring

- Gemini Vision scoring of responses
- Preserves validated scoring logic



FR-08

Cultural & Linguistic Adaptation

- Translate & culturally adapt items
- Back-translation for quality



FR-09

Clinician Report Generation

- Domain-level performance summary
- Deterministic fallback when LLM unavailable



FR-10

Reliable AI Processing

- Model router across LLM providers
- Deterministic cache & retry on failure

5 NON-FUNCTIONAL REQUIREMENTS

Quality attributes

Reliability, security, and maintainability.



Container-ready

NFR-01 **Deployment Readiness**

Web, API, worker, AI runtime, and data services are separated.



Implemented controls

NFR-02 **Security**

Auth, file gates, rate limits, tenant scoping, and service secrets.



No AI in request path

NFR-03 **Responsive Requests**

Parsing and assessment run outside the HTTP request path.



Explicit work states

NFR-04 **Failure Handling**

Retries and durable work states expose processing failures.



Cited assessment trail

NFR-05 **Explainability**

Scores retain evidence, gaps, and verification state.



Modular platform

NFR-06 **Modularity**

Separate web apps, API, worker, and inference boundary.

6 PROPOSED SOLUTION

Wayloom.AI — AI-Powered Cognitive Testing Platform



Institution foundation
Wayloom Core



Test publishing & scoring review
Wayloom Clinician Suite



Conversational testing & results
Wayloom Patient Portal

INSTITUTION ADMIN

Wayloom Core

CLINICIAN

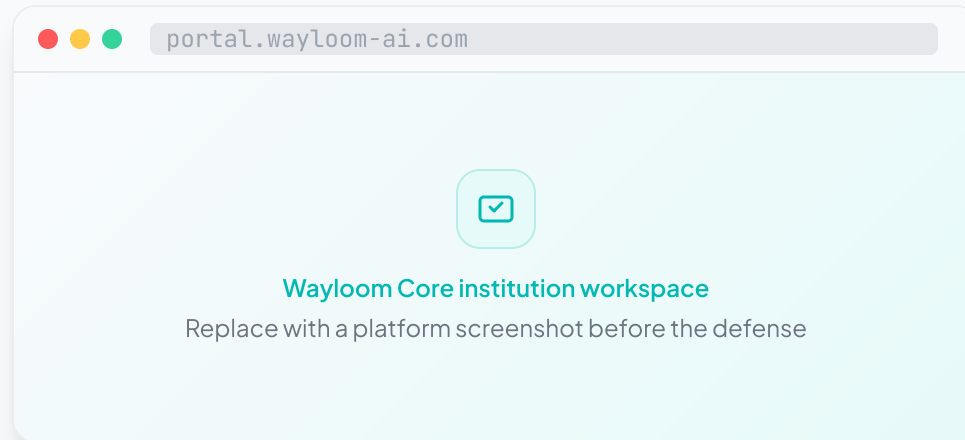
Wayloom Clinician Suite

PATIENT / CAREGIVER

Wayloom Patient Portal

6 PROPOSED SOLUTION

Wayloom Core



Wayloom Core — access, institution context, and plan controls



Identity & Access



Plans & Entitlements

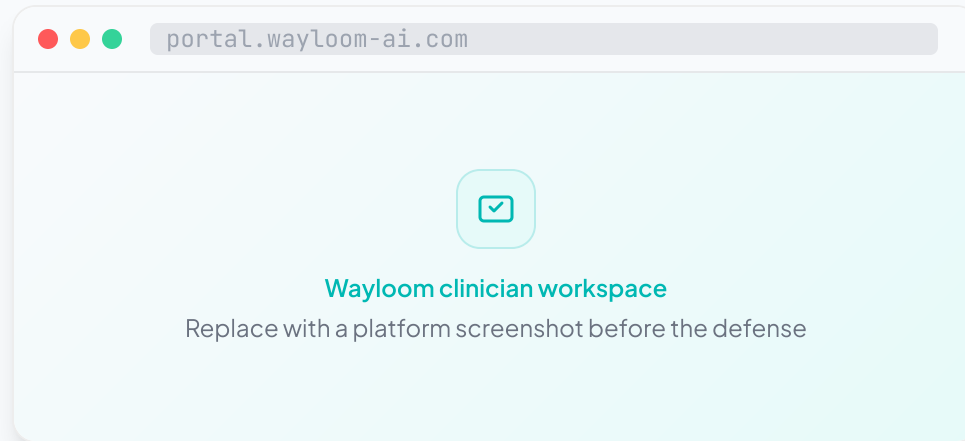


Institution Context

One control centre for institutions, clinicians, and patients

6 PROPOSED SOLUTION

Wayloom Clinician Suite



Wayloom Clinician Suite — test publishing and scoring review



Clinician test workspace



AI-assisted scoring review

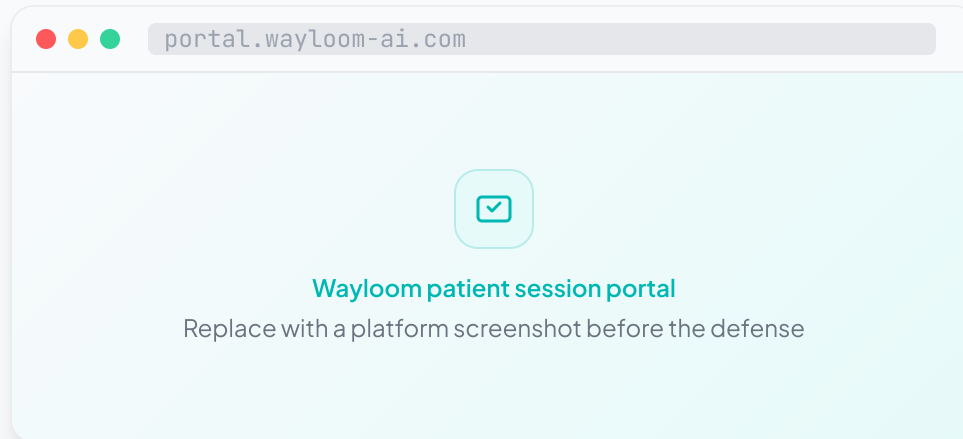


Async digitization processing

| Publish test → AI scores → clinician reviews and signs off

6 PROPOSED SOLUTION

Wayloom Patient Portal



Wayloom Patient Portal — consent, conversational testing, and results



Consent & Intake



Conversational Testing



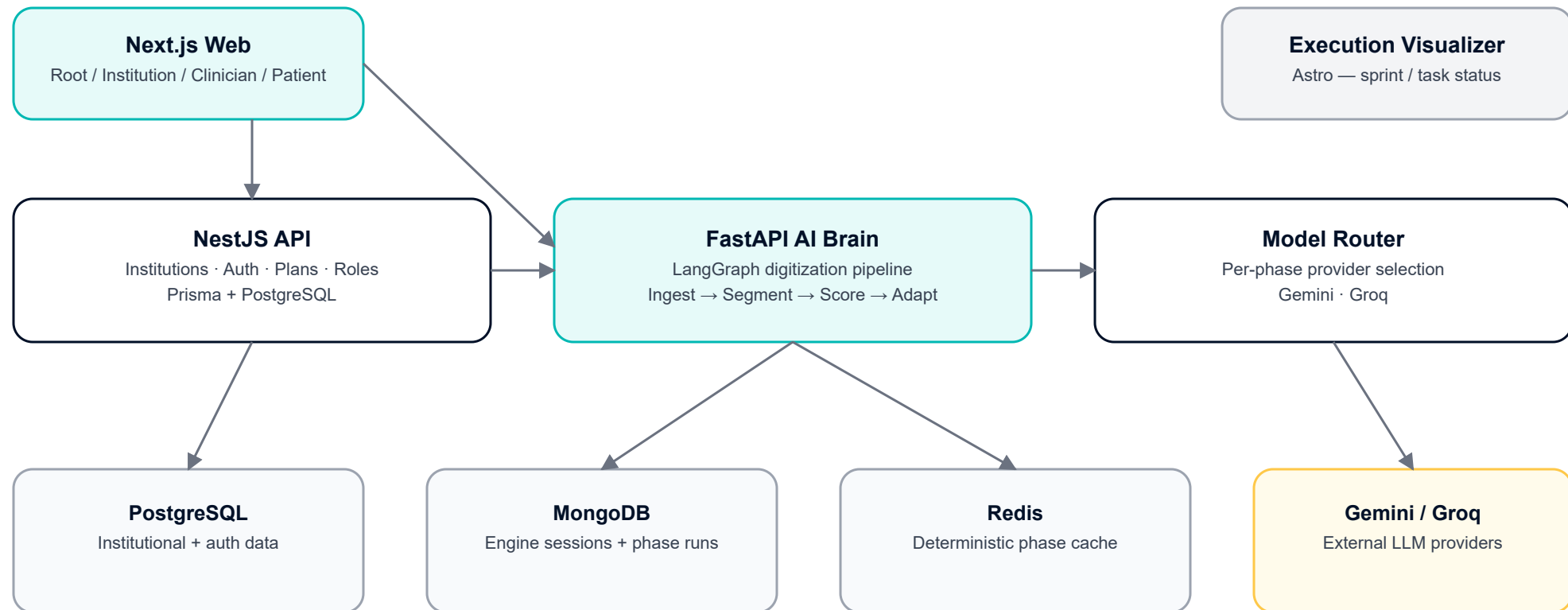
Results & Caregiver Access

| Consent → complete test → clinician & caregiver see results

7 ARCHITECTURE & TECHNOLOGIES

Logical Architecture

Client, platform, AI, and data boundaries. Gemini or Groq is selected per phase by the model router.

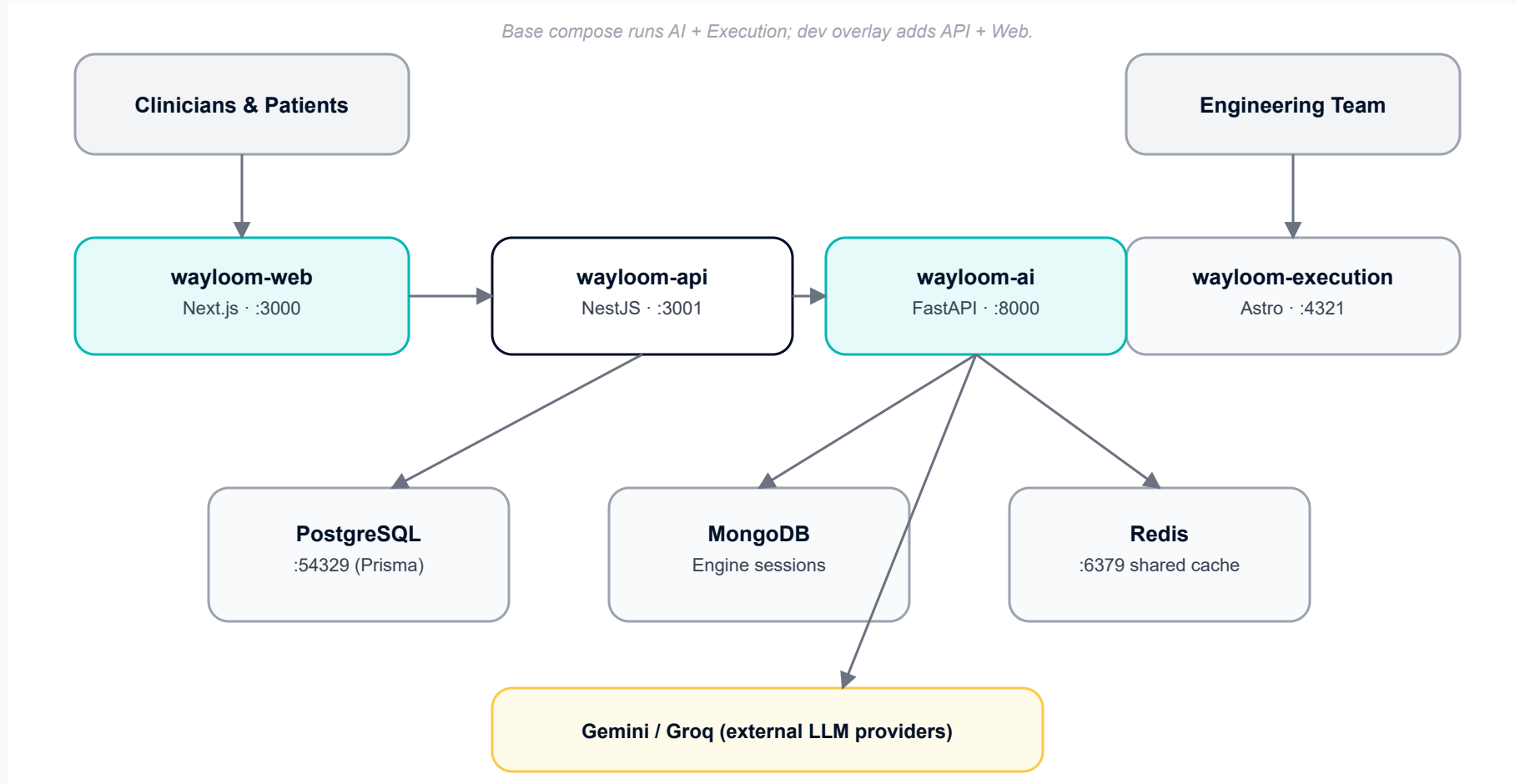


Per-app CI validates all four applications independently.

7 ARCHITECTURE & TECHNOLOGIES

Physical Architecture

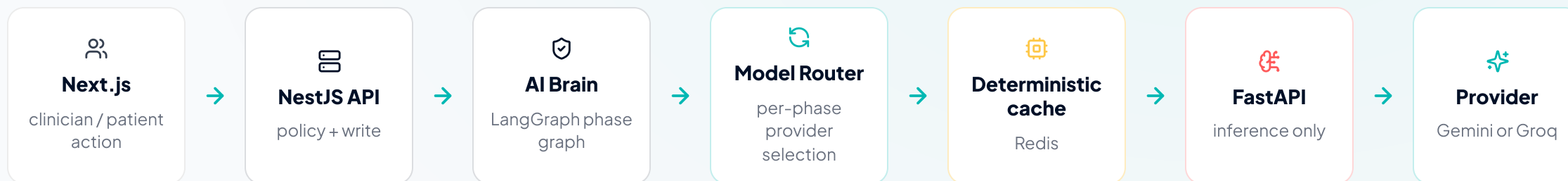
Deployment topology for web, API, AI inference, execution visualizer, and storage. Gemini or Groq is selected per phase.



8 ENGINEERING DEEP DIVE

AI Runtime Architecture

NestJS orchestrates institutional state, the AI Brain infers, and PostgreSQL/MongoDB are the source of truth.



Synchronous request path

Validate, authorize, persist institutional state, and return a processing response for short helper calls.

Digitization pipeline path

Ingest, layout, reading order, segment, score, assemble, validate, and repair — each phase cached and retryable.

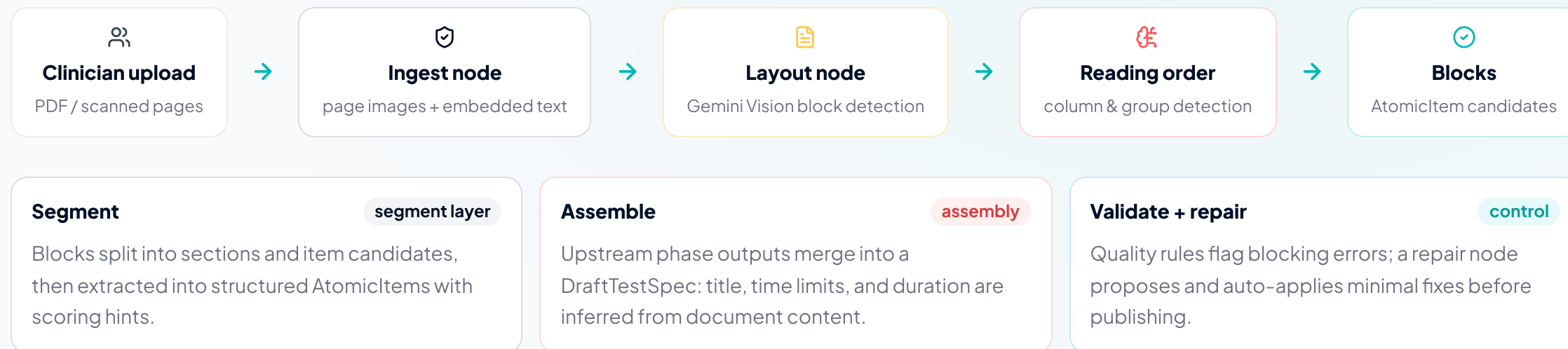
Return path: the engine writes phase results to MongoDB and PostgreSQL; Redis holds the deterministic phase cache—not authoritative state.

8 ENGINEERING DEEP DIVE

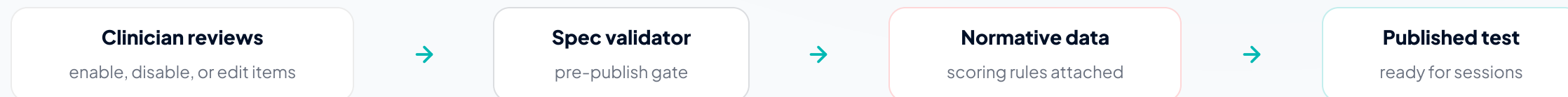
Document Digitization Pipeline

The AI Brain turns a paper test into a structured, scorable spec — without blocking the clinician.

1. INGEST, LAYOUT, AND READING ORDER

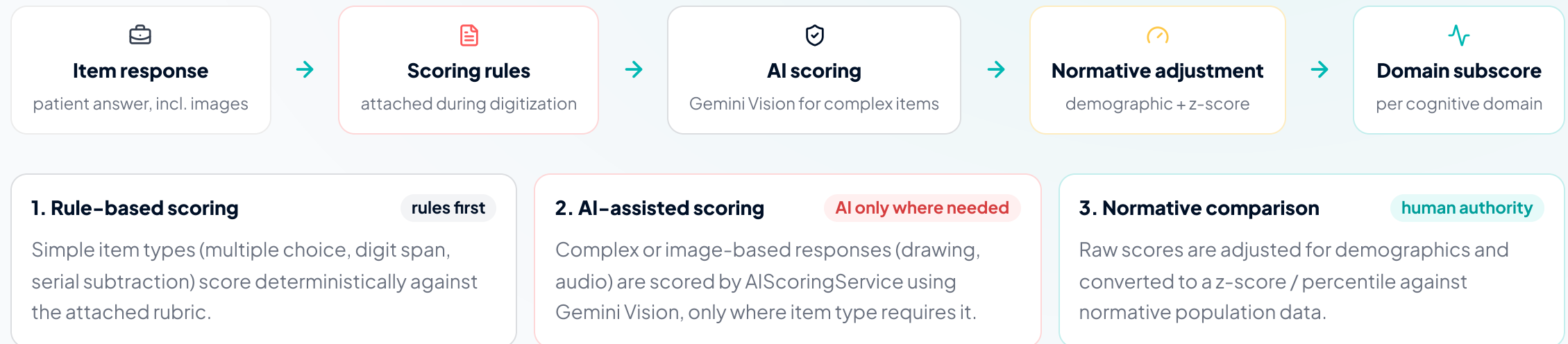


2. CLINICIAN REVIEW AND PUBLISH



AI-Assisted Scoring Engine

Deterministic scoring rules first; AI interprets complex responses; the clinician always owns the final report.



The clinician report is generated from the same persisted scores the clinician can review — not a separate opaque model output.

How Scoring Is Computed

A rule-first pipeline; AI scores only where a rubric can't, and normative data anchors the result.



1. Attach scoring rules at digitization

Scoring rules and normative data are defined by the clinician when a test is published — searched, discovered, or entered manually via the Normative Discovery Service.



2. Score at session close

Simple items score against the rubric directly. Items requiring judgment (drawing, audio, free response) are scored by Gemini Vision, constrained to the item's scoring rule.

3. Adjust against normative data

Raw score

per-item + domain totals

Z-score

vs. normative population

Demographic adjustment

age, education, culture

Domain subscore

e.g. fluency, memory, attention

4. Persist the reviewable result

Session results, domain subscores, and item-level records persist together so the clinician report always traces back to a specific response.

SCORING MODEL

domain subscore = normalize(raw score, demographic adjustment, normative z-score)

Validated scoring logic is preserved end to end — AI assists interpretation, it does not replace the rubric.

rule-based items

AI-scored items

normative lookup

AI Models by Use Case

Gemini and Groq handle inference by phase; deterministic rules and normative data anchor the score.

Use case	Technology	Role
Document layout & OCR	Gemini Vision	Page understanding, block detection
Item scoring	Gemini Vision	Score complex/image-based responses
Translation & adaptation	Groq	Cultural & linguistic adaptation, back-translation
Clinician report	Gemini	Domain-level performance summary
Model routing	Model Router	Per-phase provider selection & fallback

- Multi-provider routing
- FastAPI isolates inference
- AI does not replace the clinician's rubric

8 ENGINEERING DEEP DIVE

Performance Optimizations

Deterministic caching, phased execution, and bounded AI calls reduce latency and cost.



Deterministic cache

Redis reuse of phase outputs; never the source of truth.



Phase pipeline

Slow inference is split into cacheable, retryable phases outside the request path.



Bounded scoring

AI scoring is called only for items whose rubric requires judgment.



Adversarial test suite

Engine resilience validated against edge-case fixtures before rollout.



Provider retries

The model router retries transient provider/network failures across Gemini and Groq.



Graceful degradation

A deterministic fallback report generates when the LLM provider is unavailable.

Known limit: the Mongo → Postgres migration is not yet complete, so persistence-dependent AI endpoints still depend on MongoDB.

Technical Challenges & Solutions

Safeguards and limitations remain explicit for technical review.

Challenge	Solution	Outcome
AI latency	Phased pipeline + deterministic cache + retries	Responsive digitization path
Document quality	Vision layout + validator/repair nodes	Publishable spec or flagged for review
Cultural validity	Cultural & linguistic adaptation + back-translation	Clinically valid across language/culture
Scoring safety	Forbidden-vocab guard on patient-facing text	No unsafe or non-clinical language reaches patients
Provider outage	Model router retries + deterministic report fallback	Recoverable, observable processing
Known migration gap	Mongo → Postgres migration in progress	Documented before production

9 ROADMAP

Roadmap & Future Work

Next: harden the delivered system before widening AI capability.

NEAR-TERM



HIPAA infrastructure

Q1: build HIPAA-grade infrastructure ahead of clinical pilots.



IRB expansion

Q1: expand IRB approval alongside engine finalization.



Pilot launch

Q2: launch pilots, collect first patient data, close 3 LOIs.

MID-TERM



Peer review submission

Q3: submit clinical validation results for peer review.



Pilot expansion

Q3: expand pilot sites and file a patent.



Mongo → Postgres migration

Complete the approved persistence migration for standalone AI endpoints.

LONG-TERM VISION



Scale readiness

Q4: license agreements, deployment-ready platform, seed raise.



EHR integration pathway

Q4: scope EHR integration and Medicare/insurance reimbursement pathways.

10 RESULTS & CONCLUSION

Results & Conclusion



Engineering Achievements

- Full-stack cognitive-testing platform, IRB-approved and clinically validated



Architectural Achievements

- Four-service architecture with a deterministic, cacheable AI digitization pipeline



Lessons Learned

- Clinician authority stays central; AI assists, never replaces the rubric



Validated scoring logic, not opaque AI — early detection gives families time.

Thank you Questions?

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